

# Covariance-Method Claims Are Ensemble-Relative: Named Measures and an Audit Program

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## Abstract

A claim about the behavior of a covariance or correlation method is a claim about a distribution over matrices, and the distribution is rarely stated. We measure how much that omission matters. Across fifteen named, seeded ensembles at identical dimension and sample size, the Ledoit–Wolf shrinkage intensity ranges from 0.07 to 0.86, and eleven audited claims from eleven fields flip verdict as the ensemble changes, for stated structural reasons. The `randomcov` library names the measures and seeds the draws, so any claim can be re-run under all fifteen; the results assemble into a matrix of claims against measures, and a joint leaderboard of scikit-learn and `precise` estimators has five different winners. The pattern survives scale: six of seven claims re-run from  $n = 30$  to  $n = 3000$  keep their verdict, and the seventh, hierarchical risk parity, flips under its own home-turf ensemble as dimension grows. Eleven audits are completed here; a docket from astronomy, genomics, neuroscience, and chemometrics, including ensemble-invariant null controls, is assembled for the rest.

## 1 The unstated measure

Empirical claims about covariance methods have the form “estimator  $A$  beats estimator  $B$ ,” “the shrinkage intensity is typically moderate,” “the factorization recovers structure,” with performance averaged over random test matrices. The average is taken with respect to some generating law, and the law is usually inherited from convenience, and from the field. Finance validates on equity panels; econophysics on the same panels with random-matrix nulls; geo-statistics on smooth kernels; psychometrics on low-rank factor models; graphical modeling on sparse precisions; cosmology on mock catalogs of ordered correlation-function bins.

Different laws produce matrices with different spectra, different locality, and different factor content. A verdict certified under one law carries no warrant under another. The measurements below show that the omission is not a technicality.

## 2 Precedents

That method comparisons inherit their verdicts from the simulation design is established in several literatures. Morris et al. [2019] require simulation studies to declare the data-generating

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mechanism as a design choice of equal rank with the estimand. Boulesteix et al. [2013] trace the over-optimism of self-reported comparisons to test beds chosen by the method’s own authors. Nießl et al. [2022] re-analyze one benchmark under hundreds of design and analysis options and find that almost any method can reach almost any rank. Hooker [1995] made the complaint for heuristics; Wolpert and Macready [1997] give the limit statement, that averaged over all objectives all optimizers tie.

The correlation-matrix version also has precedent. Joe [2006] and Lewandowski et al. [2009] supply the standard samplers. Hardin et al. [2013] argue that the habitual generators produce unrealistic matrices and bias the evaluation of statistical techniques, and propose a better one. Marti [2020] learns a generator from financial correlation matrices and reports, in accompanying experiments, that the HRP advantage fades on matrices without hierarchical structure.<sup>1</sup> Papenbrock et al. [2021] evolve synthetic correlations to ask which market regimes favor which allocator. Alvarez et al. [2014] document the Bayesian mirror image: posterior conclusions about a covariance track the prior family placed on it.

The present paper adds the instrument and the audits: a fixed battery of fifteen named, seeded, public measures; the re-execution of eleven published claims from eleven fields under all of them; the claims-by-measures matrix as the reporting format; and provably invariant theorems run as null controls.

### 3 Fifteen named ensembles

`randomcov` generates correlation matrices under named, seeded measures: `lkj`, `onion`, `vine` (elliptope samplers; Joe, 2006, Lewandowski et al., 2009); `wishart`, `spectrum` (spectral laws, including Marchenko–Pastur and spiked); `factor`, `hierarchical`, `block_equicorr` (low-rank and clustered structure); `ar1`, `kernel` (banded and locality-driven fields); `sparse_precision` (graphical-model structure); `archakov_hansen` (log-matrix parameterization; Archakov and Hansen, 2021); `residuals`, `walk`, `animals` (empirical-flavored and trajectory ensembles).

Variance vectors are drawn separately, so correlation structure and scale can be crossed. Every generator accepts an explicit seed, and every number in this paper descends from one. The scripts sit beside the paper.

### 4 Demonstration: one estimator, twelve-fold spread

We begin by looking at the well known Ledoit-Wolf shrinkage estimator because this is used across so many applications. Moreover the diagnostic is easy to understand and states how far the sample covariance should be pulled towards a “target” (using neutral terminology – the target can take different flavors). The table below demonstrates the variation. Here we are fixing  $n = 30$  variables each observed exactly  $T = 60$  times. Even though this is not pressing up against the critical  $n = T$  boundary we still see tremendous variation. We used 20 draws to get a quick sense.

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<sup>1</sup>[gmarti.gitlab.io/qfin/2019/12/04/hierarchical-risk-parity-part-3.html](https://gmarti.gitlab.io/qfin/2019/12/04/hierarchical-risk-parity-part-3.html)

| ensemble         | intensity | sd    |
|------------------|-----------|-------|
| animals          | 0.072     | 0.013 |
| residuals        | 0.096     | 0.012 |
| factor           | 0.101     | 0.038 |
| archakov_hansen  | 0.120     | 0.022 |
| kernel           | 0.132     | 0.082 |
| walk             | 0.199     | 0.038 |
| block_equicorr   | 0.247     | 0.096 |
| lkj              | 0.346     | 0.031 |
| vine             | 0.355     | 0.022 |
| onion            | 0.363     | 0.030 |
| hierarchical     | 0.369     | 0.070 |
| ar1              | 0.469     | 0.226 |
| wishart          | 0.506     | 0.032 |
| spectrum         | 0.532     | 0.058 |
| sparse_precision | 0.862     | 0.066 |

The intensity spans 0.072 to 0.862 at identical  $(n, T)$ . So clearly, any paper reporting “typical” shrinkage from one generator is actually reporting a property of its generator. The ordering is interpretable too. We see that strong low-rank structure (**animals**, **residuals**, **factor**) wants little shrinkage, and near-unstructured draws (**wishart**, **sparse\_precision**) want a great deal. The within-ensemble standard deviation is itself informative. Note that **ar1** spans 0.23 across its own parameter draws.

## 5 Demonstration: one algorithm, one failing ensemble

We pick on one of our own examples. A deterministic algorithm for multinomial probit choice probabilities (the **winning** package; paper in that repository) reports median absolute error by ensemble after fitting a structured covariance grammar. Often this is quite good. In fact for twelve of the fifteen measures the error is  $1\text{--}4 \times 10^{-5}$ . On the other hand for what we term dense-strong ensembles it is  $1\text{--}4 \times 10^{-4}$  and under **kernel** fields (Matérn and RBF) it is  $4 \times 10^{-3}$ . In this case this is predictable because the fit in question does not yet admit band-diagonal structure catering to kernel covariance patterns. The point is that a validation run on any single ensemble would have reported either “essentially exact” or “inadequate,” both correctly, for the same method. Writing the random covariance generation family was a big part of the motivation: producing better software.

## 6 Eleven audits, eleven fields

Curiosity led us to wonder if the same brittleness of claim might apply more broadly, or whether, to put a more positive spin on the topic, there may be many opportunities to identify improvements to existing tools. Our methodology is very simple and short here. We take a published claim, re-run its demonstration using every type of covariance generator at our disposal, and report the verdict. The goal is not to “debunk” any claims which are usually made within their domain and subject to their own data or generation limitations. Indeed if anything the selection was biased towards things we either knew to be helpful or hoped would be.

We tried to spread this out and audit one claim from each of eleven different fields, seven about estimators and constructions (Table 1) and four about inference and dimensionality (Table 2). This part of the selection was rather ad hoc, as noted. Nonetheless Figure 1 assembles the results into a single matrix and outlines each claim’s *home turf*, by which we mean the correlation structure it was demonstrated on or intended for as best we interpret. All runs use  $n = 30$  with twenty replications unless stated. Sampled panels use  $T = 60$  except where the claim’s own regime dictates otherwise.

*Finance.* López de Prado [2016] reports that hierarchical risk parity delivers lower out-of-sample variance than quadratic minimum variance, and names estimation error in the inverse as the culprit. We build both portfolios from the same sample covariance and score the true variance  $w^\top \Sigma w$  under the generating covariance. Column 1 of Table 1 gives the median ratio of HRP to long-only minimum variance. HRP wins on every draw under `sparse_precision` and splits draws roughly evenly under `ar1` and `hierarchical`, the ensembles whose dependency structure the bisection tree can match; it loses by an order of magnitude or more under `animals`, `factor`, and `archakov_hansen`, where the true matrix rewards positions a long-only tree cannot take. Against unconstrained minimum variance the losses grow by many more orders of magnitude: under `walk` the median ratio is  $\sim 10^{8.7}$ , because near-singular ensembles admit near-perfect hedges that only signed weights reach.

*Econophysics.* Laloux et al. [1999] flatten the eigenvalues of the sample correlation that fall below the Marchenko–Pastur edge  $(1 + \sqrt{n/T})^2$  and report a cleaner matrix. Column 2 gives the Frobenius-loss ratio of clipped to raw sample correlation. We observe that this “cleaning” is excellent under `sparse_precision` (0.30) and `hierarchical` (0.76), where the true spectrum really is spikes over a bulk. On the other hand it appears to damage the estimate under eleven of fifteen ensembles, worst, at 1.43–1.47, under `animals`, `residuals`, `kernel`, and `archakov_hansen`, where genuine eigenvalue mass lives below the edge and is erased as noise. The recipe presumes signal means spikes. However in contrast, locality puts signal in the bulk. Picking the right tool for the right task is important.

*Geostatistics.* The screening effect of Stein [2002] says that a few nearest neighbors, here the most correlated, nearly attain the full kriging variance. This is a population property, so no sampling is involved. Column 3 reports the median ratio of conditional variance given the best five predictors to conditional variance given all  $n - 1$ . Screening holds essentially exactly under `ar1` (Markov, ratio 1.00), `sparse_precision`, `hierarchical`, and `block_equicorr`. It fails several-fold under `onion` and `vine`, and under `kernel` itself (5.7). Here the ensemble is mixing Matérn with squared-exponential fields and for the smooth draws interpolation is nearly deterministic from all sites, but not from five. Thus we see that Stein’s own caveat, that screening depends on smoothness, plays out.

*Psychometrics.* Horn [1965] retains the components whose sample eigenvalues exceed the corresponding 95th-percentile eigenvalue of independent data of the same shape; parallel analysis is the standard answer to “how many factors?” Column 4 gives the mean retained count: 1.8 under `sparse_precision` to 7.3 under `onion` at identical  $(n, T)$ . The same sweep scores the eigenvalue-greater-than-one rule of Kaiser [1960] against the population count of eigenvalues above one. Kaiser over-extracts under `factor` (5.7 retained against 3.0 true), under-extracts

under **wishart** (11.2 against 13.1), and disagrees with parallel analysis by a factor that is itself ensemble-dependent: 1.2 under **residuals**, 7 under **sparse\_precision**. The number of factors a study reports is, in part, a property of the generating measure its data resemble.

*Data assimilation.* Ensemble Kalman filtering runs member-starved, so to speak, and the standard way of dealing with this is covariance localization. In other words we taper the sample covariance entrywise with a Gaspari–Cohn function [Gaspari and Cohn, 1999] of inter-variable distance [Houtekamer and Mitchell, 2001, Hamill et al., 2001]. Similarly, Paz and Sánchez [2015] arrived at very much the same approach for correlation-function bins, so this audit applies to both fields. With  $N = 20$  members for  $n = 30$  variables and a taper of index distance with half-radius 6, column 5 reports the Frobenius-loss ratio of tapered to raw. The split is interpretable. Localization helps wherever the estimate is noisy and the truth carries no long-range structure in taper coordinates (including the dense ellipsope samplers) where variance reduction beats the bias. However this actually hurts under **factor**, **residuals**, and **animals** (1.66–1.71), whose common factors are long-range by construction. It also reduces performance under **kernel** (1.46), the one ensemble that genuinely is local, because kernel locality lives in a latent two-dimensional point cloud rather than in index order. Localization requires locality in the coordinates the taper can see.

*Signal processing.* Diagonal loading is the standard repair for the sample-covariance MVDR beamformer in the snapshot-limited regime [Cox et al., 1987, Carlson, 1988]. With  $T = 40$  snapshots, a random unit steering vector, the distortionless constraint, and loading  $\delta = 0.1 \text{tr}(\hat{\Sigma})/n$ , column 6 reports the median ratio of true output power, loaded over unloaded. Loading halves the output power under half the ensembles. It multiplies the power under **archakov\_hansen** (21), **animals** (33), **kernel** (136), and **walk** ( $10^{7.9}$ ). The mechanism is the hedge from the finance audit seen from the other side: under near-singular ensembles the beamformer’s null-steering lives in the smallest eigenvalues, which is exactly what loading erases.

*Graphical modeling.* Friedman et al. [2008] attach to the graphical lasso the promise that a sparsity prior on the precision matrix improves estimation. The same promise underwrites gene-network inference [Schäfer and Strimmer, 2005], partial-correlation connectomes [Smith et al., 2011], and sparse cosmological precision matrices [Padmanabhan et al., 2016]. Column 7 reports the Frobenius loss of the cross-validated glasso precision against the true precision, relative to inverting a Ledoit–Wolf shrinkage estimate. And indeed we see this. The sparsity prior wins clearly where precisions are genuinely sparse or banded (**sparse\_precision** 0.63, **hierarchical** 0.68, **ar1** 0.70), ties across the dense ellipsope middle. But it loses under **residuals** and **factor** (1.03–1.09) covariance families. No mystery here. Dense low-rank structure is the worst case for sparsity.

We also performed a smaller companion study. With oracle shrinkage intensities, the constant-correlation target of Ledoit and Wolf [2003] beats their scaled-identity target [Ledoit and Wolf, 2004] on every draw under **sparse\_precision** (median loss ratio 0.62) and loses on nineteen of twenty draws under **residuals**. Even the choice between two targets from the same authors is ensemble-relative!

Turning to other types of tests next, we have to switch to measuring hypothesis tests, rates and counts. See (Table 2) which we now discuss item by item.

| ensemble         | HRP/MV<br>(fin.) | clip/raw<br>(econo.) | near/full<br>(geost.) | PA count<br>(psych.) | taper/raw<br>(assim.) | load/raw<br>(sig. proc.) | gl/LW<br>(graph.) |
|------------------|------------------|----------------------|-----------------------|----------------------|-----------------------|--------------------------|-------------------|
| animals          | 8.4              | 1.43                 | 5.5                   | 3.3                  | 1.66                  | 32.8                     | 1.00              |
| arl              | 0.98             | 1.20                 | 1.00                  | 4.5                  | 0.53                  | 0.48                     | 0.70              |
| archakov_hansen  | 71.8             | 1.47                 | 131                   | 4.7                  | 1.41                  | 20.5                     | 1.00              |
| block_equicorr   | 1.09             | 1.00                 | 1.04                  | 3.0                  | 1.00                  | 0.55                     | 0.78              |
| factor           | 12.3             | 1.38                 | 1.06                  | 2.2                  | 1.69                  | 0.94                     | 1.09              |
| hierarchical     | 1.00             | 0.76                 | 1.02                  | 2.3                  | 0.82                  | 0.45                     | 0.68              |
| kernel           | 1.51             | 1.46                 | 5.7                   | 4.1                  | 1.46                  | 136                      | 1.00              |
| lkj              | 2.27             | 1.22                 | 3.1                   | 6.5                  | 0.74                  | 0.87                     | 0.97              |
| onion            | 2.83             | 1.29                 | 9.0                   | 7.3                  | 0.78                  | 3.3                      | 1.00              |
| residuals        | 5.36             | 1.44                 | 4.8                   | 4.7                  | 1.71                  | 0.62                     | 1.03              |
| sparse_precision | 0.76             | 0.30                 | 1.00                  | 1.8                  | 0.50                  | 0.48                     | 0.63              |
| spectrum         | 1.11             | 0.94                 | 1.32                  | 5.8                  | 0.66                  | 0.51                     | 0.99              |
| vine             | 2.68             | 1.29                 | 11.9                  | 7.0                  | 0.81                  | 4.1                      | 1.00              |
| walk             | 1.23             | 1.11                 | $\sim 10^{8.7}$       | 3.7                  | 1.14                  | $\sim 10^{7.9}$          | 1.00              |
| wishart          | 1.19             | 1.04                 | 1.50                  | 6.5                  | 0.66                  | 0.51                     | 0.96              |

Table 1: Seven audited claims about estimators and constructions; all entries are medians over twenty seeded draws. Column 1: out-of-sample true-variance ratio, HRP over long-only minimum variance (below one: HRP wins). Column 2: Frobenius-loss ratio, Marchenko–Pastur-clipped over raw sample correlation (below one: cleaning helps). Column 3: population conditional-variance ratio, five best predictors over all (near one: screening holds). Column 4: mean components retained by parallel analysis. Column 5: Frobenius-loss ratio, Gaspari–Cohn-tapered over raw sample covariance at  $N = 20$  members (below one: localization helps). Column 6: true beamformer output power, diagonally loaded over unloaded at  $T = 40$  snapshots (below one: loading helps). Column 7: precision-matrix Frobenius-loss ratio, cross-validated graphical lasso over inverted Ledoit–Wolf (below one: the sparsity prior wins). Each column’s verdict flips across rows.

*Evolutionary biology.* Cheverud [1996] declares two genetic covariance matrices similar when their mean response-vector correlation under common random selection gradients exceeds the null for independent random vectors. We apply the test to independent pairs drawn from the same ensemble. This is a population property with no sampling noise, so a sound test should declare similarity about 5% of the time. It does so under **residuals**, **factor**, and **archakov\_hansen** (0%), and declares independent matrices similar 100% of the time under eleven of the fifteen ensembles. Note however that under **sparse\_precision** the mean skewers correlation between independent matrices is 0.97, because near-diagonal matrices respond to any skewer with the skewer itself. This observation is not new. Indeed Rohlf [2017] raises this objection in general terms, anticipating the measurement here.

*Climatology.* North et al. [1982] declare a leading EOF trustworthy when its eigenvalue sampling error  $\lambda\sqrt{2/T}$  is smaller than the gap to the next eigenvalue. Where does this work? We score the false-reassurance rate, which is to say that among draws the rule declares separated, we look at how often the leading sample eigenvector is mostly wrong (squared alignment with the truth below one half). The rate is 0% under six ensembles, 57% under **wishart**, and 100% under **sparse\_precision**, which declares separation on only 12% of draws but is always wrong when it does. We take from this that the rule is a first-order perturbation statement

| ensemble         | skewers<br>false sim. | North rule<br>false reass. | Mantel<br>type-I | $n_D$ ratio<br>est/true |
|------------------|-----------------------|----------------------------|------------------|-------------------------|
| animals          | 10%                   | 0%                         | 40%              | 0.89                    |
| ar1              | 100%                  | 20%                        | 13%              | 0.54                    |
| archakov_hansen  | 0%                    | 9%                         | 28%              | 0.78                    |
| block_equicorr   | 100%                  | 0%                         | 15%              | 0.77                    |
| factor           | 0%                    | 0%                         | 38%              | 0.90                    |
| hierarchical     | 100%                  | 0%                         | 6%               | 0.78                    |
| kernel           | 100%                  | 3%                         | 39%              | 0.88                    |
| lkj              | 100%                  | 22%                        | 11%              | 0.58                    |
| onion            | 100%                  | 33%                        | 5%               | 0.57                    |
| residuals        | 0%                    | 0%                         | 37%              | 0.79                    |
| sparse_precision | 100%                  | 100%                       | 7%               | 0.42                    |
| spectrum         | 100%                  | 10%                        | 9%               | 0.52                    |
| vine             | 100%                  | 33%                        | 5%               | 0.55                    |
| walk             | 100%                  | 0%                         | 8%               | 0.83                    |
| wishart          | 100%                  | 57%                        | 7%               | 0.47                    |

Table 2: Four audited claims about inference. Column 1: share of independent matrix pairs that random skewers declares similar (sound:  $\approx 5\%$ ). Column 2: among draws North’s rule declares separated, share whose leading EOF is mostly wrong (squared alignment below  $\frac{1}{2}$ ). Column 3: realized type-I error of the Mantel test at nominal 5%. Column 4: estimated over true effective dimensions of a covariance matrix at  $T = 15$  (honest: near one).

and excellent for spiked spectra. However one should also expect it to be unreliable exactly where spectra are quasi-degenerate.

*Ecology.* Guillot and Rousset [2013] report that the Mantel test [Mantel, 1967] rejects independence between distance matrices at 25–55% under spatially autocorrelated fields, at nominal 5%. Our audit puts the original claim and the critique on the same footing (two independent site variables drawn with covariance  $C$  with Euclidean distance matrices and 499 label permutations). The realized type-I error is the nominal 5% under **onion** and **vine**, and 37–40% under **residuals**, **factor**, **kernel**, and **animals**. This reproduces the published inflation under locality-and-factor structure and shows calibration under dense elliptope measures. So in keeping with our theme, one could say “the Mantel test is broken” or “the Mantel test is fine” depending on which ensemble one chooses.

*Quantitative genetics.* Kirkpatrick [2009] reports effective dimension counts  $n_D = \sum_i \lambda_i / \lambda_{\max}$  of roughly one to two for published genetic covariance matrices, read as evolutionary constraint. At the low precision typical of breeding designs,  $T = 15$  here, the estimated  $n_D$  deflates the truth by an ensemble-dependent factor. A true  $n_D$  of 12.3 under **sparse\_precision** is reported as 5.2 (ratio 0.42), while **factor** and **animals** are nearly honest (0.90, 0.89) because their truth is already low-dimensional. Across all fifteen ensembles the estimate never exceeds 5.2, whatever the truth. A low reported  $n_D$  cannot by itself distinguish low-dimensional biology from a noisy estimate of diffuse structure.

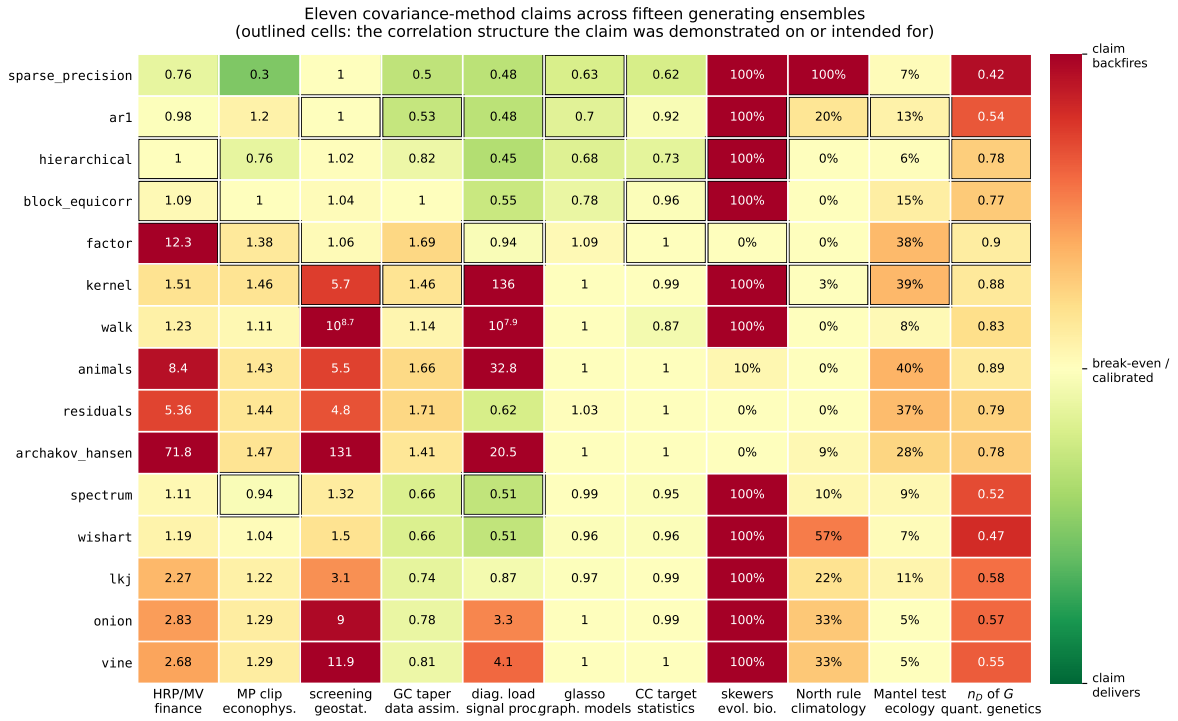


Figure 1: The audit matrix: eleven claims (columns, with their fields) against fifteen generating ensembles (rows, ordered roughly from structured to dense). Green: the claim delivers under that measure. Yellow: break-even or calibrated. Red: the claim backfires or the test is broken. Cell entries are the medians and rates of Tables 1 and 2. Outlined cells mark each claim’s home turf, the correlation structure it was demonstrated on or intended for.



## 7 Do claims scale?

Every study above fixes  $n = 30$ . A reasonable objection is that ensemble-relativity is a small-panel artifact, that estimation noise dominates at  $n = 30$  and would wash out with more assets. We re-run the Ledoit–Wolf intensity demonstration of Section 4 and five of the seven Table 1 claims (finance, econophysics, data assimilation, signal processing, and the graphical lasso, both the cross-validated penalty and a fixed one) across a  $5 \times 5$  grid,  $n \in \{30, 100, 300, 1000, 3000\}$  crossed with  $T/n \in \{0.5, 1, 2, 4, 8\}$ , medians over twenty-four seeded draws per cell (twelve at  $n = 1000$  and  $n = 3000$ ). Figure 2 plots all thirty-five cells of every audit, one line per ensemble; the two largest passes alone cost on the order of a hundred worker-hours apiece.

The spread does not close. At  $n = 3000$ ,  $T/n = 1$ , the Ledoit–Wolf intensity still ranges from 0.002 (**factor**) to 0.780 (**sparse\_precision**), a wider ratio than the 0.157–0.720 spread at  $n = 30$ . Marchenko–Pastur clipping helps **sparse\_precision** more as  $n$  grows ( $0.293 \rightarrow 0.148$  at  $T/n = 2$ ) and hurts **factor** far more ( $1.366 \rightarrow 9.432$ ); its verdict does not flip for a single literature ensemble between  $n = 30$  and  $n = 3000$  at fixed  $T/n = 2$ , and neither does diagonal loading’s, the graphical lasso’s, or Gaspari–Cohn tapering’s.<sup>2</sup> A claim’s ensemble-verdict at  $n = 30$  is, for six of the seven claims re-run, a reliable predictor of its verdict at a hundred times the dimension.

The exception is hierarchical risk parity, but only under one of the two ensembles that looked exceptional at  $n = 30$ . At  $n = 30$ ,  $T/n = 2$ , HRP roughly breaks even against long-only minimum variance under **ar1** and **hierarchical**, the two literature ensembles whose structure the bisection tree is built to match (ratios 0.890, 0.962, twenty-four draws). At  $n = 3000$ , the twelve-draw grid median moves to 1.259 and 2.580, both apparent losses. A targeted replication at forty-eight draws confirms the **hierarchical** loss (median 2.557; every draw favors minimum variance) and retracts the **ar1** one: the forty-eight-draw median is 0.914, 56% of draws still favor HRP, and the ratio ranges from 0.69 to 1.89.<sup>3</sup> The twelve-draw estimate had simply landed on the high side of a noisy statistic.

**sparse\_precision** remains an outright HRP win at every scale tested, and **ar1** remains close to a tie at every scale tested. Only **hierarchical** genuinely flips, presumably because a fixed bisection increasingly coarsens nested-cluster structure that a thirty-asset tree can still track closely, while the long-only quadratic program keeps exact use of the full covariance regardless of  $n$ . A claim’s home-turf verdict is not guaranteed to survive scale. A low rep count is not guaranteed to survive a check.

Five of the fifteen ensembles turn numerically rank-deficient at the largest dimensions tested and need a positive-semidefinite floor before any audit runs: three package-native curiosities (**walk** on 58% of  $n = 3000$  panels, **residuals** on 33%, **animals** on 17%) and two literature ensembles (**archakov\_hansen** on 14%, **kernel** on 3%). The floor is applied before scoring, so the numbers above already reflect it; a reader citing an **archakov\_hansen** or **kernel** verdict at  $n = 3000$  should know that a seventh to a thirtieth of its draws needed rescue to stay a valid correlation matrix.

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<sup>2</sup>The fixed-penalty graphical lasso records one nominal flip, **lkj** from 1.004 to 1.000, indistinguishable from a tie. The cross-validated graphical lasso is incomplete at  $n = 1000$ , roughly half its cells missing to per-cell timeouts, and dropped entirely at  $n = 3000$ , where a single fit runs into the hour.

<sup>3</sup>**hrp\_n3000\_replication.py**, beside this paper. Reps 0–11 reuse the grid’s own deterministic seeding; reps 12–47 extend it under the identical scheme.

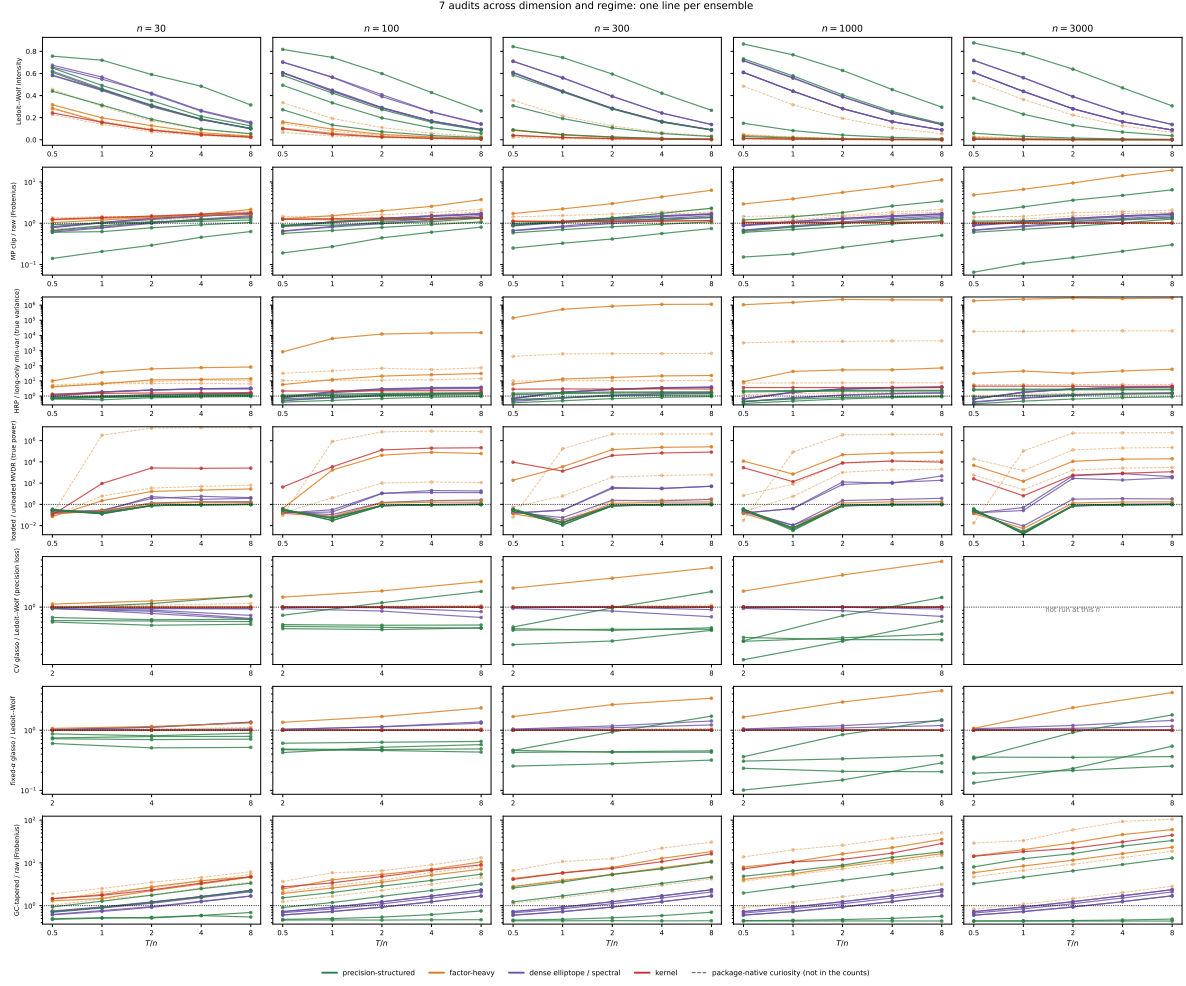


Figure 2: Seven audits (rows: Ledoit–Wolf intensity, MP clipping, HRP, MVDR loading, cross-validated and fixed-penalty graphical lasso, Gaspari–Cohn tapering) across five dimensions (columns,  $n = 30$  to  $3000$ ) and five regimes within each panel ( $T/n = 0.5$  to  $8$ ), one line per ensemble, colored by structural family; dashed lines are package-native curiosities. Rows share a  $y$ -axis scale within each audit so drift with  $n$  is visible. Cross-validated glasso is not run at  $n = 3000$ .

## 8 One leaderboard per measure

The audits test one claim at a time; the same harness ranks whole libraries. We fit the six covariance estimators of scikit-learn and the eleven batch-applicable estimators of `precise` to the same twenty seeded panels per ensemble and rank them by median relative Frobenius loss against the generating covariance (Figure 3). Five estimators take first place somewhere: Ledoit–Wolf on four ensembles, OAS and fixed shrinkage and the graphical lasso on three each, and the raw sample covariance on `animals` and `kernel`, the two ensembles the intensity table already marked as wanting almost no shrinkage. The graphical lasso is first on the three precision-structured rows and tenth or worse under `animals`, `residuals`, and `factor`. The `precise` estimators are built for time-varying covariance and pay for that design on static panels, filling the lower ranks. A library default is a bet on a measure.

## 9 Scope

We reiterate that the audits measure the ensemble-sensitivity of a claim, often outside the careful scope of the authors’ original intent. Stein proved that screening depends on smoothness; Bickel and Levina [2008] state the bandable class their banding rates require; López de Prado demonstrated HRP on equity covariances and said so; the data-assimilation literature has always tied localization to physical distance.

The interpretation risk if any usually arises downstream with less fastidious renderings of the original verdict. Those certified under one measure are quoted, cited, and defaulted into software as if they were measure-free, and the second-hand version of a scoped claim is usually unscoped. There is an obvious solution. Naming the ensembles and implementing tests can ameliorate the danger and improve understanding.

## 10 Remarks

The pattern across Figure 1 is not random: each field’s signature claim performs best on the ensembles that resemble the data the field grew up on, with instructive exceptions where a home-turf cell fails, as when screening fails inside the kernel ensemble. `sparse_precision` is kind to hierarchical bisection, eigenvalue cleaning, localization, and the graphical lasso at once, yet breaks random skewers and North’s rule. The dense elliptope samplers (`lkj`, `onion`, `vine`) are kind to almost nothing except the Mantel test, and they are the most common default test bed. The same mechanism recurs under different names: the hedge that unconstrained minimum variance exploits is the null-steering that diagonal loading erases.

A survey of prominent covariance-method claims across further fields yields a docket for the next round, kept beside this paper. Sorting it teaches a taxonomy. Claims that impose structure on the truth are ensemble-relative; theorems about Wishart sampling noise hold for every true covariance and cannot flip.

The structural claims are the audit targets. Paz and Sánchez [2015] taper clustering-statistic covariances and presume banded truth; Padmanabhan et al. [2016] estimate sparse cosmological precisions and presume a sparse graph. Zhang and Horvath [2005] select gene-network powers by a scale-free-topology criterion that is itself a statement about the generating measure of gene correlation matrices. Smith et al. [2011] demonstrated the win of partial over full correlation for network detection on sparse simulated networks. Norberg et al. [2009] judged

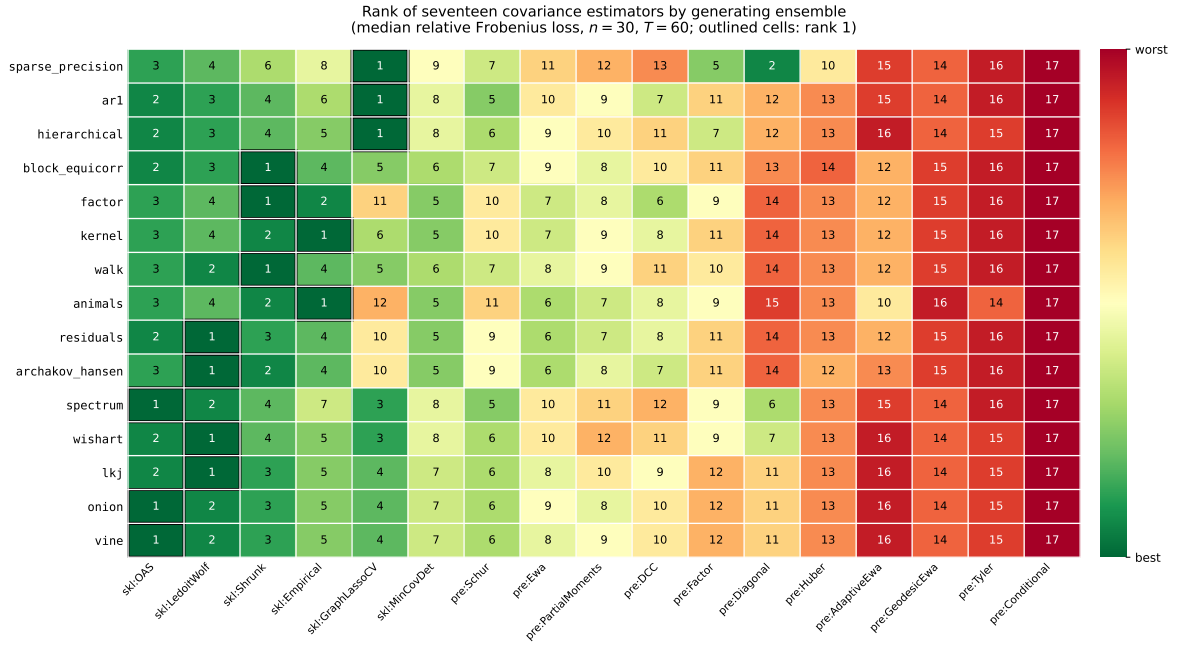


Figure 3: The leaderboard matrix: seventeen covariance estimators, six from scikit-learn and eleven from **precise**, ranked within each generating ensemble by median relative Frobenius loss at  $n = 30$ ,  $T = 60$ , twenty seeded draws per ensemble. Rank-one cells are outlined; columns are ordered by mean rank.

jackknife covariances for galaxy surveys, and the verdict hinges on cross-region correlation structure.

A second strand turns on spectral shape. Rule N [Preisendorfer and Mobley, 1988] truncates EOFs through the eigenvalue spacing law. Ledoit and Wolf [2012] derive the edge of nonlinear over linear shrinkage from spectral dispersion; Fan et al. [2013] build POET on pervasive factors plus sparse residuals; Frank and Friedman [1993] show the PLS-versus-PCR contest turning on how the coefficient vector aligns with the predictor eigenstructure. Each carries a predicted flip.

The invariant theorems serve as null controls. Hartlap et al. [2007] debias the inverse sample covariance exactly, for every true covariance; Dodelson and Schneider [2013] derive a parameter-variance inflation that depends only on counts; Sellentin and Heavens [2016] marginalize the estimated covariance exactly under Wishart sampling; Lowry et al. [1992] prove the MEWMA chart depends on a known covariance only through a noncentrality parameter. An audit pipeline that finds ensemble-dependence in the Hartlap factor has a bug, so we ran that control: with one fixed covariance per ensemble and four hundred panels at  $(n, T) = (30, 60)$ , the trace of the raw average inverse sample covariance overshoots the true inverse trace by a factor of 2.07 to 2.13 under every ensemble, against the Wishart factor  $(T - 1)/(T - n - 2) = 2.11$ , and the corrected ratio is 0.98 to 1.01 everywhere. The pipeline reports no ensemble-dependence where none can exist. The most instructive failures are invariant theorems wrapped in ensemble-relative plug-in practice, as when those corrections are applied to shrunk or tapered estimates whose distribution is no longer Wishart.

There is no free lunch. No method in the tables dominates across the rows. Each owns the ensembles whose structure it assumes, and the matrix draws the borders. We leave to future work the remaining entries, beginning with the Hartlap factor as a pipeline control, the nonlinear-against-linear shrinkage comparison, and the WGCNA scale-free criterion.

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